CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 15: Construction of Polygons

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. How many measurements are needed to construct (i) a parallelogram (ii) a rhombus (iii) a square?

2. Quadrilateral ABCD has $AB = 5$ cm, $BC = 4$ cm, $CD = 6$ cm, $DA = 4.5$ cm and $BD = 7$ cm. Which triangle is drawn first?

3. A parallelogram has diagonals 10 cm and 6 cm meeting at 60°. What lengths are marked from the point of intersection?

4. A rectangle has a side 8 cm and a diagonal 10 cm. What is the other side?

5. A rhombus has diagonals 10 cm and 24 cm. What will its side measure?

6. A square has a diagonal 6 cm. What will its side measure, to one decimal place?

7. In trapezium ABCD with $AB \parallel DC$, $AB = 8$ cm and $DC = 5$ cm. CE is drawn parallel to DA to meet AB at E. Find EB.

8. What is the radius of the circle used to construct a regular hexagon of side 4 cm?

9. Can a quadrilateral have sides 3 cm, 4 cm, 5 cm and 13 cm?

10. Which compass angles are combined to construct 105°?

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Answer Key

1. 3, 2 and 1
2. Triangle ABD (or BCD), on the diagonal BD
3. 5 cm on each side along one diagonal and 3 cm on each side along the other
4. 6 cm
5. 13 cm
6. 4.2 cm
7. 3 cm
8. 4 cm
9. No; the longest side must be less than the sum of the other three, and $3 + 4 + 5 = 12 < 13$
10. 90° and 15° (or 60° and 45°)

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